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Apparatuses and methods for securely presenting digital objects

Abstract

Apparatuses, methods, and computer program products are provided for securely presenting digital objects. An example method includes receiving a request for digital object review and causing visual presentation of the digital object to a user. The method further includes receiving one or more images of the user's eye captured during presentation of the digital object and determining a focus location represented by the one or more images. The method also includes modifying one or more user parameters associated with the user based on the focus location. In some instances, presentation of the digital object may be rendered in a virtual reality environment or may be overlaid as an augmented reality object in the field of view of the user.

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Background/Summary

CROSS REFERENCE TO RELATED APPLICATIONS (1) The present application is a continuation application of U.S. patent application Ser. No. 16/900,256, filed Jun. 12, 2020, which is hereby incorporated by reference in its entirety.

TECHNOLOGICAL FIELD

(1) Example embodiments of the present disclosure relate generally to customer insights and, more particularly, to the use of securely presented digital objects to improve customer interactions.

BACKGROUND

(2) Many institutions, entities, businesses, and the like often require that a customer review, sign, or otherwise acknowledge various forms in order to perform an action with these entities. For example, in order for a user to open an account or perform a transaction with a financial institution, the user may be required to review and complete various documents. In many instances, a user may also be required to visit a physical location in order to complete this process.

BRIEF SUMMARY

(3) As described above, financial institutions and other entities may utilize a variety documents or forms in the normal course of interacting with their customers. Customers may, for example, enter a physical branch of a financial institution and review numerous documents in order to open an account with this institution. Given the volume of text presented to the customer, in many instances, customers may be reluctant to provides questions regarding the content of the document or may fail to review portions of the documents. Customers are also becoming increasingly concerned with the privacy of their personal information and may be wary of the use of physical documents that may be misplace or viewed by others. Furthermore, the options provided to a customer at a physical location may be limited due to the availability of select employees (e.g., those tasked with handling a particular action), due to technology limitations (e.g., particular equipment only located at select locations), or the like.

(4) To solve these issues and others, example implementations of embodiments of the present disclosure may leverage virtual reality (VR) and/or augmented realty (AR) to provide a secure environment for customer interactions. In operation, embodiments of the present disclosure may receive a request for digital object review (e.g., documents or the like) and may cause visual presentation of the digital object to the user in a VR or AR environment. During review of the digital object in the VR or AR environment, the system may capture images of the user's eye to determine one or more focus locations of the user with respect to the digital object. As such, the system may operate to determine or otherwise map locations of interest of the digital object (e.g., associated with increased focus of the user) while protecting private user information in the VR or AR environment (e.g., only viewable by the user). In this way, the inventors have identified that the advent of emerging computing technologies have created a new opportunity for solutions for securely presenting digital content and determining customer insights which were historically unavailable. In doing so, such example implementations confront and solve at least two technical challenges: (1) they reliably secure user data from unauthorized viewing, and (2) they determine

customer insights in a virtual environment (e.g., without the need for physical forms).

(5) As such, apparatuses, methods, and computer program products are provided for securely presenting digital objects. With reference to an example method, the example method may include receiving, by a computing device, a request for digital object review and causing, by display circuitry of the computing device, visual presentation of the digital object to a user. The method may further include receiving, by gaze detection circuitry, one or more images of the user's eye captured during presentation of the digital object and determining, by the gaze detection circuitry, a focus location represented by the one or more images. The method may also include modifying, by object analysis circuitry of the computing device, one or more user parameters associated with the user based on the focus location.

(6) In some embodiments, causing presentation of the digital object to the user may further include generating, via virtual reality circuitry of the computing device, a virtual reality environment. In such an embodiment, the method may include rendering, via the virtual reality circuitry, the digital object in the virtual reality environment.

(7) In other embodiments, causing presentation of the digital object to the user may further include overlaying, via augmented reality circuitry of the computing device, the digital object in a field of view of the user.

(8) In some embodiments, the method may further include augmenting, via the object analysis circuitry, the digital object based on the one or more parameters of the user. In such an embodiment, augmenting the digital object may further include generating, via the object analysis circuitry, a visual representation of the focus location on the digital object.

(9) In some embodiments, the method may also include transmitting, via the computing device, a notification to the user comprising one or more modified user parameters.

(10) In some embodiments, determining the focus location may further include identifying, via the gaze detection circuitry, a gaze direction of the user's eye. The method may further include comparing, via the gaze detection circuitry, the gaze direction of the user's eye with a corresponding location defined by the digital object and determining, via the gaze detection circuitry, the focus location based upon a correlation between the gaze direction and the corresponding location of the digital object.

(11) The above summary is provided merely for purposes of summarizing some example embodiments to provide a basic understanding of some aspects of the invention. Accordingly, it will be appreciated that the above-described embodiments are merely examples and should not be construed to narrow the scope or spirit of the invention in any way. It will be appreciated that the scope of the invention encompasses many potential embodiments in addition to those here summarized, some of which will be further described below.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

(1) Having described certain example embodiments of the present disclosure in general terms above, reference will now be made to the accompanying drawings. The components illustrated in the figures may or may not be present in certain embodiments described herein. Some embodiments may include fewer (or more) components than those shown in the figures.

(2) FIG. 1 illustrates a system diagram including devices that may be involved in some example embodiments described herein.

(3) FIG. 2 illustrates a schematic block diagram of example circuitry that may perform various operations, in accordance with some example embodiments described herein.

(4) FIG. 3 illustrates an example flowchart for securely presenting digital objects, in accordance with some example embodiments described herein.

(5) FIG. 4 illustrates an example flowchart for virtual reality (VR) and augmented reality (AR) digital object presentation, in accordance with some example embodiments described herein.

(6) FIG. 5 illustrates an example flowchart for focus location determinations, in accordance with some example embodiments described herein.

DETAILED DESCRIPTION

(7) Some embodiments of the present disclosure will now be described more fully hereinafter with reference to the accompanying drawings, in which some, but not all embodiments are shown. Indeed, this disclosure may be embodied in many different forms and should not be construed as limited to the embodiments set forth herein; rather, these embodiments are provided so that this disclosure will satisfy applicable legal requirements. Like numbers refer to like elements throughout. As used herein, the description may refer to a digital presentation server as an example “apparatus.” However, elements of the apparatus described herein may be equally applicable to the claimed method and computer program product. Thus, use of any such terms should not be taken to limit the spirit and scope of embodiments of the present disclosure.

DEFINITION OF TERMS

(8) As used herein, the terms “data,” “content,” “information,” “electronic information,” “signal,” “command,” and similar terms may be used interchangeably to refer to data capable of being transmitted, received, and/or stored in accordance with embodiments of the present disclosure. Thus, use of any such terms should not be taken to limit the spirit or scope of embodiments of the present disclosure. Further, where a first computing device is described herein to receive data from a second computing device, it will be appreciated that the data may be received directly from the second computing device or may be received indirectly via one or more intermediary computing devices, such as, for example, one or more servers, relays, routers, network access points, base stations, hosts, and/or the like, sometimes referred to herein as a “network.” Similarly, where a first computing device is described herein as sending data to a second computing device, it will be appreciated that the data may be sent directly to the second computing device or may be sent indirectly via one or more intermediary computing devices, such as, for example, one or more servers, remote servers, cloud-based servers (e.g., cloud utilities), relays, routers, network access points, base stations, hosts, and/or the like.

(9) As used herein, the term “comprising” means including but not limited to and should be interpreted in the manner it is typically used in the patent context. Use of broader terms such as comprises, includes, and having should be understood to provide support for narrower terms such as consisting of, consisting essentially of, and comprised substantially of.

(10) As used herein, the phrases “in one embodiment,” “according to one embodiment,” “in some embodiments,” and the like generally refer to the fact that the particular feature, structure, or characteristic following the phrase may be included in at least one embodiment of the present disclosure. Thus, the particular feature, structure, or characteristic may be included in more than one embodiment of the present disclosure such that these phrases do not necessarily refer to the same embodiment.

(11) As used herein, the word “example” is used herein to mean “serving as an example, instance, or illustration.” Any implementation described herein as “example” is not necessarily to be construed as preferred or advantageous over other implementations.

(12) As used herein, the terms “user device,” “first user device,” “second user device,” and the like refer to computer hardware and/or software that is configured to access a service made available by the digital presentation server and, among various other functions, is configured to directly, or indirectly, transmit and receive data. Example user devices may include a smartphone, a tablet computer, a laptop computer, a wearable device (e.g., smart glasses, smartwatch, or the like), virtual reality (VR) headset, augmented reality (AR) device and the like. In some embodiments, a user device may include a “smart device” that is equipped with chip of other electronic device that is configured to communicate with the digital presentation server via Bluetooth, NFC, Wi-Fi, 3G,

4G, 5G, RFID protocols, and the like. By way of a particular example, a first user device may be an virtual or augmented reality device (e.g., VR headset, smart glasses, etc.) associated with a first user (e.g., customer), and a second user device may be a virtual or augmented reality device (e.g., VR headset, smart glasses, etc.) associated with a second user (e.g., financial advisory or other banking employee), where each user device is equipped with a Wi-Fi radio that is configured to communicate with a Wi-Fi access point that is in communication with the a server (e.g., a digital presentation server of the present disclosure) via a network.

(13) As used herein, the term “user parameters” may refer to a collection of settings, configurations, identifiers, data, and information associated with a user and associated user device. User parameters configured in accordance with the present disclosure may be stored by an associated user device, user profile, or the like and may be accessible by one or more of the software applications that are supported by the digital presentation server and, thus, may include application-specific preferences, settings, configurations, data, and information. In some example embodiments, user parameters of a first user may include account information, preferences, transaction information, and/or the like associated with the first user as described hereafter.

(14) As used herein, the terms “virtual reality” or “VR” may refer to a user simulation or simulated experience in which a user's physical presence is provided in a virtual environment. Such a virtual environment may include any number of artificial items, features, or the like with which the user may interact. Furthermore, a VR environment may include various forms of feedback to a user's interactions (e.g., audio or tactile feedback) and may utilize associated VR headsets, head-mounted displays, or multi-projected environments. Similarly, the terms “augmented reality” or “AR” may refer to an interactive experience that enhances objects in a real-world environment. Said differently, AR may combine real and virtual worlds or items and, in some instances, may operate as an overlay of a user's real environment. As opposed to virtual reality in which a user's environment is partially or completely simulated, an augmented reality environment may only project virtual items into a user's field of view. An AR environment may utilize associated AR headsets, smart glasses, displays, or the like in order to enhance the user's real world environment.

(15) As used herein, the term “computer-readable medium” refers to non-transitory storage hardware, non-transitory storage device or non-transitory computer system memory that may be accessed by a controller, a microcontroller, a computational system or a module of a computational system to encode thereon computer-executable instructions or software programs. A non-transitory “computer-readable medium” may be accessed by a computational system or a module of a computational system to retrieve and/or execute the computer-executable instructions or software programs encoded on the medium. Exemplary non-transitory computer-readable media may include, but are not limited to, one or more types of hardware memory, non-transitory tangible media (for example, one or more magnetic storage disks, one or more optical disks, one or more USB flash drives), computer system memory or random access memory (such as, DRAM, SRAM, EDO RAM), and the like.

(16) Having set forth a series of definitions called-upon throughout this application, an example system architecture and example apparatus is described below for implementing example embodiments and features of the present disclosure.

Device Architecture and Example Apparatus

(17) With reference to FIG. 1, an example system **100** is illustrated with an apparatus (e.g., a digital presentation server **200**) communicably connected via a network **104** to a first user device **102**, and, in some embodiments, a second user device **106**. Although illustrated connected to the digital presentation server **200** via a network **104**, the present disclosure contemplates that one or more of the first user device **102** and/or the second user device **106** may be hosted and/or stored by the digital presentation server **200**.

(18) The digital presentation server **200** may include circuitry, networked processors, or the like configured to perform some or all of the apparatus-based (e.g., digital presentation server-based)

processes described herein, and may be any suitable network server and/or other type of processing device. In this regard, digital presentation server **200** may be embodied by any of a variety of devices. For example, the digital presentation server **200** may be configured to receive/transmit data and may include any of a variety of fixed terminals, such as a server, desktop, or kiosk, or it may comprise any of a variety of mobile terminals, such as a portable digital assistant (PDA), mobile telephone, smartphone, laptop computer, tablet computer, or in some embodiments, a peripheral device that connects to one or more fixed or mobile terminals. Example embodiments contemplated herein may have various form factors and designs but will nevertheless include at least the components illustrated in FIG. 2 and described in connection therewith. In some embodiments, the digital presentation server **200** may be located remotely from the first user device **102** and/or the second user device **106**, although in other embodiments, the digital presentation server **200** may comprise the first user device **102** and/or the second user device **106**. The digital presentation server **200** may, in some embodiments, comprise several servers or computing devices performing interconnected and/or distributed functions. Despite the many arrangements contemplated herein, the digital presentation server **200** is shown and described herein as a single computing device to avoid unnecessarily overcomplicating the disclosure.

(19) The network **104** may include one or more wired and/or wireless communication networks including, for example, a wired or wireless local area network (LAN), personal area network (PAN), metropolitan area network (MAN), wide area network (WAN), or the like, as well as any hardware, software and/or firmware for implementing the one or more networks (e.g., network routers, switches, hubs, etc.). For example, the network **104** may include a cellular telephone, mobile broadband, long term evolution (LTE), GSM/EDGE, UMTS/HSPA, IEEE 802.11, IEEE 802.16, IEEE 802.20, Wi-Fi, dial-up, and/or WiMAX network. Furthermore, the network **104** may include a public network, such as the Internet, a private network, such as an intranet, or combinations thereof, and may utilize a variety of networking protocols now available or later developed including, but not limited to TCP/IP based networking protocols.

(20) The first user device **102** may be associated with a first user (e.g., customer) and may be configured to store or access various user parameters associated with the first user. Although a single first user device **102** is shown in FIG. 1, the example system **100** may include any number of user devices associated with the first user. The first user device **102** may be a cellular telephone (including smartphones and/or other types of mobile telephones), laptop, tablet, electronic reader, e-book device, media device, wearable, smart glasses, smartwatch, VR device, AR device, or any combination of the above. The first user device **102** may be communicably coupled with the digital presentation server **200**, via the network **104**, and configured to transmit data to and receive data from the digital presentation server **200**. By way of example, the first user device **102** may include VR or AR device configured to generate a VR environment or AR overlay, respectively, for viewing by the first user.

(21) In some embodiments, the second user device **106** may be associated with a second user (e.g., employee of financial institution) and may be configured to store or access various user parameters associated with the second user. Although a single second user device **106** is shown in FIG. 1, the example system **100** may include any number of user devices associated with the second user. The second user device **106** may be a cellular telephone (including smartphones and/or other types of mobile telephones), laptop, tablet, electronic reader, e-book device, media device, wearable, smart glasses, smartwatch, VR device, AR device, or any combination of the above. The second user device **106** may be communicably coupled with the digital presentation server **200**, via the network **104**, and configured to transmit data to and receive data from the digital presentation server **200**. By way of example, the second user device **106** may include VR or AR device configured to generate a VR environment or AR overlay, respectively, for viewing by the second user.

(22) As illustrated in FIG. 2, the digital presentation server **200** may include a processor **202**, a memory **204**, communications circuitry **208**, and input/output circuitry **206**. Moreover, the digital

presentation server **200** may include gaze detection circuitry **210**, object analysis circuitry **214**, and/or display circuitry **212**. The digital presentation server **200** may be configured to execute the operations described below in connection with FIGS. 3-5. Although components **202-218** are described in some cases using functional language, it should be understood that the particular implementations necessarily include the use of particular hardware. It should also be understood that certain of these components **202-218** may include similar or common hardware. For example, two sets of circuitry may both leverage use of the same processor **202**, memory **204**, communications circuitry **208**, or the like to perform their associated functions, such that duplicate hardware is not required for each set of circuitry. The use of the term “circuitry” as used herein includes particular hardware configured to perform the functions associated with respective circuitry described herein. As described in the example above, in some embodiments, various elements or components of the circuitry of the digital presentation server **200** may be housed within the first user device **102**, and/or the second user device **106**. It will be understood in this regard that some of the components described in connection with the digital presentation server **200** may be housed within one of these devices, while other components are housed within another of these devices, or by yet another device not expressly illustrated in FIG. 1.

(23) Of course, while the term “circuitry” should be understood broadly to include hardware, in some embodiments, the term “circuitry” may also include software for configuring the hardware. For example, although “circuitry” may include processing circuitry, storage media, network interfaces, input/output devices, and the like, other elements of the digital presentation server **200** may provide or supplement the functionality of particular circuitry.

(24) In some embodiments, the processor **202** (and/or co-processor or any other processing circuitry assisting or otherwise associated with the processor) may be in communication with the memory **204** via a bus for passing information among components of the digital presentation server **200**. The memory **204** may be non-transitory and may include, for example, one or more volatile and/or non-volatile memories. In other words, for example, the memory may be an electronic storage device (e.g., a non-transitory computer readable storage medium). The memory **204** may be configured to store information, data, content, applications, instructions, or the like, for enabling the digital presentation server **200** to carry out various functions in accordance with example embodiments of the present disclosure.

(25) The processor **202** may be embodied in a number of different ways and may, for example, include one or more processing devices configured to perform independently. Additionally, or alternatively, the processor may include one or more processors configured in tandem via a bus to enable independent execution of instructions, pipelining, and/or multithreading. The use of the term “processing circuitry” may be understood to include a single core processor, a multi-core processor, multiple processors internal to the digital presentation server, and/or remote or “cloud” processors.

(26) In an example embodiment, the processor **202** may be configured to execute instructions stored in the memory **204** or otherwise accessible to the processor **202**. Alternatively, or additionally, the processor **202** may be configured to execute hard-coded functionality. As such, whether configured by hardware or by a combination of hardware with software, the processor **202** may represent an entity (e.g., physically embodied in circuitry) capable of performing operations according to an embodiment of the present disclosure while configured accordingly. Alternatively, as another example, when the processor **202** is embodied as an executor of software instructions, the instructions may specifically configure the processor **202** to perform the algorithms and/or operations described herein when the instructions are executed.

(27) The digital presentation server **200** further includes input/output circuitry **206** that may, in turn, be in communication with processor **202** to provide output to a user and to receive input from a user, user device, or another source. In this regard, the input/output circuitry **206** may comprise a display that may be manipulated by a mobile application. In some embodiments, the input/output circuitry **206** may also include additional functionality such as a keyboard, a mouse, a joystick, a

touch screen, touch areas, soft keys, a microphone, a speaker, or other input/output mechanisms. The processor **202** and/or user interface circuitry comprising the processor **202** may be configured to control one or more functions of a display through computer program instructions (e.g., software and/or firmware) stored on a memory accessible to the processor (e.g., memory **204**, and/or the like).

(28) The communications circuitry **208** may be any means such as a device or circuitry embodied in either hardware or a combination of hardware and software that is configured to receive and/or transmit data from/to a network and/or any other device, circuitry, or module in communication with the digital presentation server **200**. In this regard, the communications circuitry **208** may include, for example, a network interface for enabling communications with a wired or wireless communication network. For example, the communications circuitry **208** may include one or more network interface cards, antennae, buses, switches, routers, modems, and supporting hardware and/or software, or any other device suitable for enabling communications via a network. Additionally, or alternatively, the communication interface may include the circuitry for interacting with the antenna(s) to cause transmission of signals via the antenna(s) or to handle receipt of signals received via the antenna(s). These signals may be transmitted by the digital presentation server **200** using any of a number of wireless personal area network (PAN) technologies, such as Bluetooth® v1.0 through v3.0, Bluetooth Low Energy (BLE), infrared wireless (e.g., IrDA), ultra-wideband (UWB), induction wireless transmission, or the like. In addition, it should be understood that these signals may be transmitted using Wi-Fi, Near Field Communications (NFC), Worldwide Interoperability for Microwave Access (WiMAX) or other proximity-based communications protocols.

(29) The gaze detection circuitry **210** includes hardware components designed to capture one or more images of a user's eye during review of a digital object and analyze said images. In particular, the gaze detection circuitry may identify a gaze direction of a user's eye and compare this gaze direction with a corresponding location defined by a digital object. The gaze detection circuitry **210** may utilize processing circuitry, such as the processor **202**, to perform its corresponding operations, and may utilize memory **204** to store collected information.

(30) The display circuitry **212** includes hardware components designed to cause visual presentation of a digital object to a user. In some embodiments, the display circuitry **212** may further leverage virtual reality circuitry **216** to generate a virtual reality environment and render the digital object in the virtual reality environment. Similarly, in some embodiments, the display circuitry **212** may further leverage augmented reality circuitry **218** to overlay the digital object in the field of view of the user (e.g., in a real world environment). The display circuitry **212**, the virtual reality circuitry **216**, and/or the augmented reality circuitry **218** may utilize processing circuitry, such as the processor **202**, to perform its corresponding operations, and may utilize memory **204** to store collected information.

(31) The object analysis circuitry **214** includes hardware components designed to modify one or more user parameters associated with a user based on a determined focus location. In some embodiments, the object analysis circuitry **214** may generate a visual representation of the focus location on the digital object. The object analysis circuitry **214** may utilize processing circuitry, such as the processor **202**, to perform its corresponding operations, and may utilize memory **204** to store collected information.

(32) It should also be appreciated that, in some embodiments, the gaze detection circuitry **210**, display circuitry **212**, object analysis circuitry **214**, virtual reality circuitry **216**, and/or augmented reality circuitry **218** may include a separate processor, specially configured field programmable gate array (FPGA), or application specific interface circuit (ASIC) to perform its corresponding functions.

(33) In addition, computer program instructions and/or other type of code may be loaded onto a computer, processor, or other programmable digital presentation server's circuitry to produce a

machine, such that the computer, processor or other programmable circuitry that execute the code on the machine create the means for implementing the various functions, including those described in connection with the components of digital presentation server **200**.

(34) As described above and as will be appreciated based on this disclosure, embodiments of the present disclosure may be configured as systems, methods, mobile devices, and the like.

Accordingly, embodiments may comprise various means including entirely of hardware or any combination of software with hardware. Furthermore, embodiments may take the form of a computer program product comprising instructions stored on at least one non-transitory computer-readable storage medium (e.g., computer software stored on a hardware device). Any suitable computer-readable storage medium may be utilized including non-transitory hard disks, CD-ROMs, flash memory, optical storage devices, or magnetic storage devices.

Example Operations for Securely Presenting Digital Objects

(35) FIG. 3 illustrates a flowchart containing a series of operations for generating user connections. The operations illustrated in FIG. 3 may, for example, be performed by, with the assistance of, and/or under the control of an apparatus (e.g., digital presentation server **200**), as described above. In this regard, performance of the operations may invoke one or more of processor **202**, memory **204**, input/output circuitry **206**, communications circuitry **208**, gaze detection circuitry **210**, display circuitry **212**, object analysis circuitry **214**, virtual reality circuitry **216**, and/or augmented reality circuitry **218**.

(36) As shown in operation **305**, the apparatus (e.g., digital presentation server **200**) includes means, such as input/output circuitry **206**, communications circuitry **208**, or the like, for receiving a request for digital object review. As described above, the first user device **102** may be associated with a first user and may, in some embodiments transmit the request for digital object review at operation **305**. In other embodiments, the digital presentation server **200** may be configured to periodically transmit or ping connected user devices and attempt to cause presentation of a virtual or augmented reality. Said differently, the digital presentation server **200** may periodically attempt to identify user devices that are communicably coupled to the server **200**, if any, and offer to generate a VR or AR environment as described herein. By way of example, the digital presentation server **200** may be associated with a branch location of a financial institution and may identify user devices connected to a network of the branch location.

(37) In some embodiments, the request received at operation **305** may be transmitted by a second user device **106**. By way of example, the second user device **106** may be associated with a second user that is an employee of the bank branch. As such, the employee (e.g., financial advisor or the like) may use a second user device **106** associated with the second user to transmit the request to the digital presentation server **200**. By way of a particular example, a customer (e.g., a first user) may enter a physical branch location and request assistance with opening an account, applying for a loan, or the like and may be paired with an employee of the bank branch (e.g., a second user). In order to view the various documents required for opening the account, the customer via a first user device **102** (e.g., VR or AR headset) and/or the employee via a second device **106** (e.g., VR or AR headset) may transmit a request to perform the review of these documents (e.g., digital objects) in a VR or AR environment.

(38) Thereafter, as shown in operation **310**, the apparatus (e.g., digital presentation server **200**) includes means, such as input/output circuitry **206**, communications circuitry **208**, display circuitry **212**, virtual reality circuitry **216**, augmented reality circuitry **218**, or the like, for causing visual presentation of the digital object to a user. As described hereafter with reference to FIG. 4, the display circuitry **212** may be configured to, based on the first user device **102** and/or the second user device **106**, generate a virtual reality environment or augmented reality overlay.

(39) By way of example, in an instance in which the first user device **102** and/or the second user device **106** are VR devices, the display circuitry **212** and, in particular the virtual reality circuitry **216**, may cause the first user device **102** and the second user device **106** to generate a virtual reality

environment and render the digital object (e.g., documents, forms, etc.) in the virtual reality environment. By way of continued example, the first user device **102** may be worn (e.g., VR headset) by a customer (e.g., first user) and the second user device **106** (e.g., VR device) may be worn by an employee of the banking branch. Upon receiving the request at operation **305**, the virtual reality circuitry **216** may transmit instructions to the first user device **102** and the second user device **106** to generate a virtual reality environment for simultaneous viewing by the customer and the employee. In some instances, the virtual reality environment may generate a virtual reality environment that simulates the physical banking branch. However, the present disclosure contemplates that any simulated environment may be generated by the first user device **102** and the second user device **106** (e.g., in response to instructions from the digital presentation server **200**).

(40) The display circuitry **212** and virtual reality circuitry **216** may be further configured to render the requested digital object (e.g., document, form, etc.) in the virtual reality environment. Said differently, the display circuitry **212** or virtual reality circuitry **216** may simulate the digital object in the virtual reality environment in a form that is viewable by the user. By way of example, the digital object may correspond to a disclosure document that is viewable by the user via the first user device **102** in the virtual reality environment. Similarly, in some embodiments, the second user device **106** may allow the second user to simultaneously view the disclosure document (e.g., digital object) in the same or separate virtual reality environments. Although described herein with reference to a disclosure document, form, paper, or similar digital object, the present disclosure contemplates that the digital object may include any object, element, video, chart, image, or the like without limitation. Furthermore, the present disclosure contemplates that the digital object, when rendered in a virtual reality environment, may have any form, shape, dimension, size, or the like to facilitate viewing by the first user.

(41) With continued reference to operation **310**, and as described hereafter with reference to FIG. **4**, in an instance in which the first user device **102** and/or the second user device **106** are AR devices, the display circuitry **212** and/or augmented reality circuitry **218**, may cause the first user device **102** and the second user device **106** to generate an augmented reality overlay and overlay the digital object (e.g., documents, forms, etc.) in the field of view of the user (e.g., in a real world environment). By way of continued example, the first user device **102** may be worn (e.g., AR headset, smart glasses, etc.) by a customer (e.g., first user) and the second user device **106** (e.g., AR device, smart glasses, etc.) may be worn by an employee of the banking branch. Upon receiving the request at operation **305**, the augmented reality circuitry **218** may transmit instructions to the first user device **102** and the second user device **106** to generate an augmented reality overlay that includes the digital object for simultaneous viewing by the customer (e.g., first user) and the employee (e.g., second user). In some instances, the augmented reality overlay may enhance the real world environment of the physical banking branch as viewed by the first user through the first user device **102**.

(42) The display circuitry **212** and augmented reality circuitry **218** may be further configured to overlay the digital object in a field of view of the first user. By way of example, the digital object may correspond to a disclosure document that is viewable by the user via through first user device **102** in a real world environment enhanced by the augmented reality overlay. Similarly, in some embodiments, the second user device **106** may allow the second user to simultaneously view the disclosure document (e.g., digital object) via an overlay of the real world environment. Although described herein with reference to a disclosure document, form, paper, or similar digital object, the present disclosure contemplates that the digital object may include any object, element, video, chart, image, or the like without limitation. Furthermore, the present disclosure contemplates that the digital object, when overlaid via an augmented reality overlay, may have any form, shape, dimension, size, or the like to facilitate viewing by the first user. For example, the digital object may be illustrated to the first user on a piece of paper (e.g., blank piece of paper that exists in the real world environment) such that the content of the digital object is only viewable by the first user

device **102** and/or second user device **106**.

(43) Thereafter, as shown in operation **315**, the apparatus (e.g., digital presentation server **200**) includes means, such as processor **202**, gaze detection circuitry **210**, or the like, for receiving one or more images of the user's eye captured during presentation of the digital object. In some example embodiments, and as described in greater detail below in connection with FIG. 5, operation **315** may include receiving one or more images of a user's eye via a camera (e.g., or equivalent device) of the first user device **102**. By way of example, a digital object may be presented to the user as described above with reference to operation **310** such that a context, text, etc. of the digital object is displayed, for example, as a rendering in a virtual reality environment or as an augmented reality overlay. A camera of the first user device **102** or other eye-tracking sensor may begin capturing (e.g., as a video stream, collection of images, or the like) one or more images of a user's eye. The user may begin reading or otherwise viewing the content of the digital object by directing his or her gaze to portions of the digital object (e.g., reading the text of the digital object).

(44) In some embodiments, as shown in operation **320**, the apparatus (e.g., digital presentation server **200**) includes means, such as processor **202**, gaze detection circuitry **210**, or the like, for determining a focus location represented by the one or more images. As described hereafter with reference to FIG. 5, the gaze detection circuitry **210** may iteratively determine the relative position of a user's eye (e.g., or any portion of the user's eye, pupil, etc.) relative to the digital object, user's head, or other frame of reference. This relative positioning of the user's eye may be referred to as the user's gaze direction. For each gaze direction, the gaze detection circuitry **210** may identify a corresponding location of the digital object. Said differently, the gaze detection circuitry **210** may receive a plurality of images of a user's eye and determine a gaze direction for each image as well as a corresponding location of the digital object for each gaze direction (e.g., where is the user looking?). Subsequently, the gaze detection circuitry **210** may determine a correlation between the gaze direction and the corresponding location of the digital object in order to determine a focus location.

(45) By way of example, the first user may view a disclosure document (e.g., digital object) via the first user device and may focus or otherwise direct his or her gaze (e.g., attention) to portions of the digital object. The gaze detection circuitry **210** may be configured to determine the corresponding portion of the digital object associated with the user's gaze as described with reference to FIG. 5 and may further associate a time component with this association. Said differently, the gaze detection circuitry **210** may determine an amount of time that the user's gaze is directed to a portion of the digital object. In order to determine a focus location, the gaze detection circuitry **210** may further compare the time component of each gaze direction. This comparison may be used to identify the portion of the digital object at which the user's gaze was directed for the most time (e.g., or any amount of time based on the intended application). By way of continued example, the first user may direct his or her gaze at a portion of the disclosure document (e.g., digital object) that contains content related to a minimum account balance. Said differently, the gaze detection circuitry **210** may determine the content of the portion of the digital object at the focus location for use in modifying operation of the system as described hereafter with reference to operation **325**.

(46) In some embodiments, as shown in operation **325**, the apparatus (e.g., digital presentation server **200**) includes means, such as processor **202**, object analysis circuitry **214**, or the like, for modifying one or more user parameters associated with the user based on the focus location. As described above, the focus location may be associated with the user's gaze direction and digital object and may further indicate potential concern of the user with respect to the content of the digital object at the focus location. By way of continued example, a disclosure document viewed by the user may include text relating to minimum balance requirements for opening an account. Based upon the gaze detection techniques described herein, the server **200** may determine that the focus location is associated with the portion of the digital object relating the minimum balance

requirements.

(47) In some embodiments, the modification of one or more user parameters at operation **325** may refer to a modification of parameters available for review by the second user associated with the second user device **106**. For example, the object analysis circuitry **214** may modify user parameters to indicate the first user's potential concern for minimum balance requirements. In response, the second user may review these modifications and adjust discussions with the first user accordingly. In particular, the second user may attempt to reassure the first user regarding these requirements, may attempt to provide alternative options for proceeding, may provide additional information regarding these requirements, and/or the like. This modification at operation **325** may be stored in a user profile regarding the first user or similar storage location so as to inform future interactions with the first user.

(48) In some embodiments, as shown in operation **330**, the apparatus (e.g., digital presentation server **200**) includes means, such as input/output circuitry **206**, display circuitry **212**, object analysis circuitry **214**, or the like, for augmenting the digital object based on the one or more parameters of the user. The object analysis circuitry **214** may, based on the modifications to the one or more user parameters, adjust, modify, or otherwise change the digital object. By way of continued example, the object analysis circuitry **214** may highlight (e.g., any visual indicator) this portion of the digital content for review with the first user. In some embodiments, the augmenting of the digital object at operation **330** may refer to removing a portion (e.g., the focus location) of the digital object from the digital object. Said differently, the object analysis circuitry **214** may determine the concern of the first user regarding minimum balance requirements and may select a different account opening without a minimum balance such that the digital object no longer includes this focus location (e.g., area of user concern). Although described herein with reference to a single focus location regarding minimum balance, the present disclosure contemplates that any number of focus locations may be determined by the systems described herein. Furthermore the focus locations may be associated with any content type and/or any number of associated modifications may be made to the user parameters in response to this focus location.

(49) In some embodiments, the object analysis circuitry **214** may be configured to augment the digital object by generating a visual representation of the focus location on the digital object. In such an embodiment, the object analysis circuitry **214** may operate to form a heat map or other visualization of representative magnitude of the user's gaze. By way of continued example, the disclosure document (e.g., digital object) may be augmented to include a heat map associated with the first user's gaze at each portion of the disclosure. Said differently, the digital object may be augmented to include a color coded mapping associated with the respective gaze of the user for each portion of the disclosure. Such a heat map may be viewable by the second user via the second user device **106** in order to provide tailored guidance to the first user.

(50) Turning next to FIG. **4**, a flowchart is shown for virtual reality (VR) and augmented reality (AR) digital object presentation. The operations illustrated in FIG. **4** may, for example, be performed by, with the assistance of, and/or under the control of an apparatus (e.g., digital presentation server **200**), as described above. In this regard, performance of the operations may invoke one or more of processor **202**, memory **204**, input/output circuitry **206**, communications circuitry **208**, gaze detection circuitry **210**, display circuitry **212**, object analysis circuitry **214**, virtual reality circuitry **216**, and/or augmented reality circuitry **218**.

(51) As shown in operation **405**, the apparatus (e.g., digital presentation server **200**) includes means, such as input/output circuitry **206**, communications circuitry **208**, or the like, for receiving a request for digital object review. As described above at operation **305**, the first user device **102** may be associated with a first user and may, in some embodiments transmit the request for digital object review at operation **405**. In other embodiments, the digital presentation server **200** may be configured to periodically transmit or ping connected user devices and attempt to cause presentation of a virtual or augmented reality. Said differently, the digital presentation server **200**

may periodically attempt to identify user devices that are communicably coupled to the server **200**, if any, and offer to generate a VR or AR environment as described herein. In other embodiments, the request received at operation **405** may be transmitted by a second user device **106**.

(52) As shown in operation **410**, the apparatus (e.g., digital presentation server **200**) includes means, such as input/output circuitry **206**, communications circuitry **208**, display circuitry **212**, augmented reality circuitry **218**, or the like, for overlaying the digital object in a field of view of the user. As described above with reference to operation **310**, in an instance in which the first user device **102** and/or the second user device **106** are AR devices, the display circuitry **212** and/or augmented reality circuitry **218**, may cause the first user device **102** and the second user device **106** to generate an augmented reality overlay and overlay the digital object (e.g., documents, forms, etc.) in the field of view of the user (e.g., in a real world environment). The display circuitry **212** and augmented reality circuitry **218** may be further configured to overlay the digital object in a field of view of the first user. By way of example, the digital object may correspond to a disclosure document that is viewable by the user via through first user device **102** in a real world environment enhanced by the augmented reality overlay. Similarly, in some embodiments, the second user device **106** may allow the second user to simultaneously view the disclosure document (e.g., digital object) via an overlay of the real world environment.

(53) As shown in operation **415**, the apparatus (e.g., digital presentation server **200**) includes means, such as input/output circuitry **206**, communications circuitry **208**, display circuitry **212**, augmented reality circuitry **218**, gaze detection circuitry **210**, or the like, for receiving one or more images of the user's eye captured of the overlaid digital object. As described above, one or more images of a user's eye may be captured via a camera (e.g., or equivalent device) of the first user device **102**. In particular, a camera of the first user device **102** or other eye-tracking sensor may begin capturing (e.g., as a video stream, collection of images, or the like) one or more images of a user's eye directed at the overlaid digital object. The user may begin reading or otherwise viewing the content of the digital object by directing his or her gaze to portions of the overlaid digital object (e.g., reading the text of the digital object). Such images may be used by the gaze detection circuitry **210** in determining a focus location represented by the one or more images as described above with reference to operation **320**.

(54) As shown in operations **425** and **430**, the apparatus (e.g., digital presentation server **200**) includes means, such as input/output circuitry **206**, display circuitry **212**, virtual reality circuitry **216**, or the like, for generating a virtual reality environment and rendering the digital object in the virtual reality environment. As described above with reference to operation **310**, in an instance in which the first user device **102** and/or the second user device **106** are VR devices, the display circuitry **212** and, in particular the virtual reality circuitry **216**, may cause the first user device **102** and the second user device **106** to generate a virtual reality environment and render the digital object (e.g., documents, forms, etc.) in the virtual reality environment. Upon receiving the request at operation **405**, the virtual reality circuitry **216** may transmit instructions to the first user device **102** and the second user device **106** to generate a virtual reality environment for simultaneous viewing by the customer and the employee. The display circuitry **212** and virtual reality circuitry **216** may be further configured to render the requested digital object (e.g., document, form, etc.) in the virtual reality environment. Said differently, the display circuitry **212** or virtual reality circuitry **216** may simulate the digital object in the virtual reality environment in a form that is viewable by the user.

(55) Thereafter, as shown in operation **430**, the apparatus (e.g., digital presentation server **200**) includes means, such as input/output circuitry **206**, communications circuitry **208**, display circuitry **212**, virtual reality circuitry **216**, gaze detection circuitry **210**, or the like, for receiving one or more images of the user's eye captured of the rendered digital object. As described above, one or more images of a user's eye may be captured via a camera (e.g., or equivalent device) of the first user device **102**. In particular, a camera of the first user device **102** or other eye-tracking sensor may

begin capturing (e.g., as a video stream, collection of images, or the like) one or more images of a user's eye directed at the digital object rendered in a virtual reality environment. The user may begin reading or otherwise viewing the content of the digital object by directing his or her gaze to portions of the rendered digital object (e.g., reading the text of the digital object). Such images may be used by the gaze detection circuitry **210** in determining a focus location represented by the one or more images as described above with reference to operation **320**.

(56) Turning next to FIG. 5, a flowchart is shown for focus location determinations. The operations illustrated in FIG. 5 may, for example, be performed by, with the assistance of, and/or under the control of an apparatus (e.g., digital presentation server **200**), as described above. In this regard, performance of the operations may invoke one or more of processor **202**, memory **204**, input/output circuitry **206**, communications circuitry **208**, gaze detection circuitry **210**, display circuitry **212**, object analysis circuitry **214**, virtual reality circuitry **216**, and/or augmented reality circuitry **218**.

(57) As shown in operation **505**, the apparatus (e.g., digital presentation server **200**) includes means, such as input/output circuitry **206**, gaze detection circuitry **210**, or the like, for identifying a gaze direction of the user's eye. As described above with reference to FIG. 3, the gaze detection circuitry **210** may receive one or more images of a user's eye via a camera (e.g., or equivalent device) of the user device **102**. By way of example, a camera of a mobile phone may capture (e.g., as a video stream, collection of images, or the like) one or more images of a user's eye during review of the digital object. In order to determine the gaze direction at operation **505**, the gaze detection circuitry **210** may identify a user's face, eye, pupil, and/or other related features captured in the one or more images of the user's eye. In some embodiments, the gaze detection circuitry **210** may further identify a vector between the center of the user's pupil and a reflection (e.g., corneal reflections) on the user's eye as a result of the presented digital object. This vector may be indicative of the user's gaze direction. While described herein with reference to vector techniques and corneal reflections, the present disclosure contemplates that any eye tracking or gaze detection technique (e.g., active light, dark pupil, bright pupil, passive light, and/or the like) may be used to determine the gaze direction of the user's eye. Given that a period of time may elapse during review of the digital object, the gaze direction of the user's eye may be iteratively identified over the time in which the identification code is inputted.

(58) Thereafter, as shown in operations **510**, **515** the apparatus (e.g., digital presentation server **200**) includes means, such as processor **202**, gaze detection circuitry **210**, or the like, for comparing the gaze direction of the user's eye with a corresponding location defined by the digital object and determining the focus location based upon the correlation, respectively. As would be evident to one of ordinary skill in the art in light of the present disclosure, the digital object (e.g., disclosure or other document) may define various location (e.g., text sections, paragraphs, etc.) each associated with content of the digital object. These respective locations may be known by or otherwise provided to the gaze detection circuitry **210** for comparison with the gaze direction(s) of the user's eye identified at operation **505**. The gaze detection circuitry **210** may compare the known location of each portion of the digital object with each gaze direction in order to determine one or more focus locations of the user.

(59) Thereafter, as shown in operation **520**, the apparatus (e.g., digital presentation server **200**) includes means, such as processor **202**, object analysis circuitry **214**, or the like, for modifying one or more user parameters associated with the user based on the focus location. As described above with reference to operation **325**, the focus location may be associated with the user's gaze direction and digital object and may further indicate potential concern of the user with respect to the content of the digital object at the focus location. By way of continued example, a disclosure document viewed by the user may include text relating to minimum balance requirements for opening an account. Based upon the gaze detection techniques described herein, the server **200** may determine that the focus location is associated with the portion of the digital object relating the minimum

balance requirements.

(60) In some embodiments, as shown in operations **525** the apparatus (e.g., digital presentation server **200**) includes means, such as processor **202**, communications circuitry **208**, or the like, for transmitting a notification to the user (e.g., first user device **102**) comprising one or more modified user parameters. By way of continued example, the modification of one or more user parameters at operation **520** may refer to a modification of parameters available for review by the second user associated with the second user device **106**. For example, the object analysis circuitry **214** may modify user parameters to indicate the first user's potential concern for minimum balance requirements. In response, the second user may review these modifications and adjust discussions with the first user accordingly. In particular, the second user may attempt to reassure the first user regarding these requirements, may attempt to provide alternative options for proceeding, may provide additional information regarding these requirements, and/or the like. At operation **525**, in some embodiments, the server **200** may transmit a notification to the user that includes the modification to the user parameters (e.g., modified user parameters). For example the notification may indicate to the user that assistance may be provided regarding the minimum balance requirements associated with the account opening.

(61) In doing so, the embodiments of the present disclosure solve issues regarding digital object presentation, customer insight development, and document security. The embodiments of the present disclosure leverage virtual reality (VR) and/or augmented reality (AR) to provide a secure environment for customer interactions. In operation, embodiments of the present disclosure receive a request for digital object review (e.g., documents or the like) and cause visual presentation of the digital object to the user in a VR or AR environment. During review of the digital object in the VR or AR environment, the system capture images of the user's eye to determine one or more focus locations of the user with respect to the digital object. As such, the system operates to determine or otherwise map locations of interest of the digital object (e.g., associated with increased focus of the user) while protecting private user information in the VR or AR environment (e.g., only viewable by the user). In this way, the inventors have identified that the advent of emerging computing technologies have created a new opportunity for solutions for securely presenting digital content and determining customer insights which were historically unavailable. In doing so, such example implementations confront and solve at least two technical challenges: (1) they reliably secure user data from unauthorized viewing, and (2) they determine customer insights in a virtual environment (e.g., without the need for physical forms).

(62) FIGS. **3-5** thus illustrate flowcharts describing the operation of apparatuses, methods, and computer program products according to example embodiments contemplated herein. It will be understood that each flowchart block, and combinations of flowchart blocks, may be implemented by various means, such as hardware, firmware, processor, circuitry, and/or other devices associated with execution of software including one or more computer program instructions. For example, one or more of the operations described above may be implemented by an apparatus executing computer program instructions. In this regard, the computer program instructions may be stored by a memory **204** of the digital presentation server **200** and executed by a processor **202** of the digital presentation server **200**. As will be appreciated, any such computer program instructions may be loaded onto a computer or other programmable apparatus (e.g., hardware) to produce a machine, such that the resulting computer or other programmable apparatus implements the functions specified in the flowchart blocks. These computer program instructions may also be stored in a computer-readable memory that may direct a computer or other programmable apparatus to function in a particular manner, such that the instructions stored in the computer-readable memory produce an article of manufacture, the execution of which implements the functions specified in the flowchart blocks. The computer program instructions may also be loaded onto a computer or other programmable apparatus to cause a series of operations to be performed on the computer or other programmable apparatus to produce a computer-implemented process such that the instructions

executed on the computer or other programmable apparatus provide operations for implementing the functions specified in the flowchart blocks.

(63) The flowchart blocks support combinations of means for performing the specified functions and combinations of operations for performing the specified functions. It will be understood that one or more blocks of the flowcharts, and combinations of blocks in the flowcharts, can be implemented by special purpose hardware-based computer systems which perform the specified functions, or combinations of special purpose hardware with computer instructions.

CONCLUSION

(64) Many modifications of the embodiments set forth herein will come to mind to one skilled in the art to which this disclosure pertains having the benefit of the teachings presented in the foregoing descriptions and the associated drawings. Therefore, it is to be understood that the disclosure is not to be limited to the specific embodiments disclosed and that modifications and other embodiments are intended to be included within the scope of the appended claims. Moreover, although the foregoing descriptions and the associated drawings describe example embodiments in the context of certain example combinations of elements and/or functions, it should be appreciated that different combinations of elements and/or functions may be provided by alternative embodiments without departing from the scope of the appended claims. In this regard, for example, different combinations of elements and/or functions than those explicitly described above are also contemplated as may be set forth in some of the appended claims. Although specific terms are employed herein, they are used in a generic and descriptive sense only and not for purposes of limitation.

Claims

1. A method for securely presenting digital objects, the method comprising: receiving, by communications circuitry, a request for digital object review; rendering, by virtual reality circuitry, a digital object in at least a first virtual reality environment and a second virtual reality environment that is different than the first virtual reality environment in response to the request for digital object review, wherein the digital object is a digital representation of a document; causing, by display circuitry, visual presentation of the digital object within the first virtual reality environment to a first user via a first virtual reality device and simultaneously within the second virtual reality environment to a second user via a second virtual reality device; receiving, by gaze detection circuitry, one or more images of an eye of the first user captured during presentation of the digital object; determining, by the gaze detection circuitry, a focus location with respect to the digital object; generating, by object analysis circuitry, a visual representation of the focus location on the digital object, wherein (i) the visual representation of the focus location is visually representative of a magnitude of a gaze of the first user with respect to the digital object and (ii) the visual representation of the focus location on the digital object is solely viewable by the second user via the second virtual reality device; determining, by the object analysis circuitry and based on the focus location, a concern of the first user with respect to one or more portions of content of the digital object at the focus location, wherein the concern of the first user is associated with at least a first portion of text comprised in the one or more portions of content of the digital object at the focus location; modifying, by the object analysis circuitry, one or more user parameters associated with the first user based on the focus location and the concern of the first user; generating, by the object analysis circuitry and based on modifying the one or more user parameters, one or more modified user parameters, wherein the one or more modified user parameters associated with the first user are viewable by a second user via a second virtual reality device; updating, by the object analysis circuitry and based on the one or more modified user parameters, account information associated with the first user; and transmitting, by the communications circuitry, a notification to the second virtual reality device comprising the one or more modified user parameters, wherein the

notification indicates the concern of the first user.

2. The method of claim 1, the method further comprising: identifying, by the gaze detection circuitry, a gaze direction of the eye of the first user; comparing, by the gaze detection circuitry, the gaze direction of the eye of the first user with a corresponding location defined by the digital object; determining, by the gaze detection circuitry, the focus location based upon a correlation between the gaze direction and the corresponding location of the digital object; and determining, by the object analysis circuitry, the concern of the first user based on the focus location and the gaze direction.

3. The method of claim 1, further comprising augmenting, by the object analysis circuitry, at least a first portion of content of the one or more portions of content of the digital object based on the focus location and the concern of the first user, wherein augmenting the first portion of content of the digital object comprises one or more of (i) visually indicating the first portion of the content of the digital object corresponding to the focus location, (ii) removing the first portion of the content of the digital object corresponding to the focus location, (iii) replacing the first portion of the content of the digital object corresponding to the focus location, or (iv) generating a new digital object that does not include the first portion of content of the digital object corresponding to the focus location.

4. The method of claim 1, wherein the one or more user parameters are associated with a user profile of the first user, and wherein the one or more user parameters comprise at least one or more of the account information, one or more user preferences, one or more concerns of the first user, one or more user identifiers, one or more user device identifiers, or transaction information associated with the first user.

5. An apparatus for securely presenting digital objects, the apparatus comprising: communications circuitry configured to receive a request for digital object review; virtual reality circuitry configured to render a digital object in at least a first virtual reality environment and a second virtual reality environment that is different than the first virtual reality environment in response to the request for digital object review, wherein the digital object is a digital representation of a document; display circuitry configured to cause visual presentation of the digital object within the first virtual reality environment to a first user via a first virtual reality device and simultaneously within the second virtual reality environment to a second user via a second virtual reality device; gaze detection circuitry configured to: receive one or more images of an eye of the first user captured during presentation of the digital object, and determine a focus location with respect to the digital object; and object analysis circuitry configured to: generate a visual representation of the focus location on the digital object, wherein (i) the visual representation of the focus location is visually representative of a magnitude of a gaze of the first user with respect to the digital object and (ii) the visual representation of the focus location on the digital object is solely viewable by the second user via the second virtual reality device; determine, based on the focus location, a concern of the first user with respect to one or more portions of content of the digital object at the focus location, wherein the concern of the first user is associated with at least a first portion of text comprised in the one or more portions of content of the digital object at the focus location; modify one or more user parameters associated with the first user based on the focus location and the concern of the first user; generate, based on modifying the one or more user parameters, one or more modified user parameters, wherein the one or more modified user parameters associated with the first user are viewable by a second virtual reality device; and update, based on the one or more modified user parameters, account information associated with the first user, wherein the communications circuitry is further configured to transmit a notification to the second virtual reality device comprising the one or more modified user parameters, and wherein the notification indicates the concern of the first user.

6. The apparatus of claim 5, wherein the gaze detection circuitry is further configured to: identify a gaze direction of the eye of the first user; compare the gaze direction of the eye of the first user

with a corresponding location defined by the digital object; and determine the focus location based upon a correlation between the gaze direction and the corresponding location of the digital object, wherein the object analysis circuitry is further configured to determine the concern of the first user based on the focus location and the gaze direction.

7. The apparatus of claim 5, wherein the object analysis circuitry is further configured to augment at least a first portion of content of the one or more portions of content of the digital object based on the focus location and the concern of the first user, wherein augmenting the first portion of content of the digital object comprises one or more of (i) visually indicating the first portion of the content of the digital object corresponding to the focus location, (ii) removing the first portion of the content of the digital object corresponding to the focus location, (iii) replacing the first portion of the content of the digital object corresponding to the focus location, or (iv) generating a new digital object that does not include the first portion of content of the digital object corresponding to the focus location.

8. The apparatus of claim 5, wherein the one or more user parameters are associated with a user profile of the first user, and wherein the one or more user parameters comprise at least one or more of the account information, one or more user preferences, one or more concerns of the first user, one or more user identifiers, one or more user device identifiers, or transaction information associated with the first user.

9. A non-transitory computer-readable storage medium for using an apparatus for securely presenting digital objects, the non-transitory computer-readable storage medium storing instructions that, when executed, cause the apparatus to: receive a request for digital object review; render a digital object in at least a first virtual reality environment and a second virtual reality environment that is different than the first virtual reality environment in response to the request for digital object review, wherein the digital object is a digital representation of a document; cause visual presentation of the digital object within the first virtual reality environment to a first user via a first virtual reality device and simultaneously within the second virtual reality environment to a second user via a second virtual reality device; receive one or more images of an eye of the first user captured during presentation of the digital object; determine a focus location with respect to the digital object; generate a visual representation of the focus location on the digital object, wherein (i) the visual representation of the focus location is visually representative of a magnitude of a gaze of the first user with respect to the digital object and (ii) the visual representation of the focus location on the digital object is solely viewable by the second user via the second virtual reality device; determine, based on the focus location, a concern of the first user with respect to one or more portions of content of the digital object at the focus location, wherein the concern of the first user is associated with at least a first portion of text comprised in the one or more portions of content of the digital object at the focus location; modify one or more user parameters associated with the first user based on the focus location and the concern of the first user; generate, based on modifying the one or more user parameters, one or more modified user parameters, wherein the one or more modified user parameters associated with the first user are viewable by a second user via a second virtual reality device; update, based on the one or more modified user parameters, account information associated with the first user; and transmit, a notification to the second virtual reality device comprising the one or more modified user parameters, wherein the notification indicates the concern of the first user.

10. The non-transitory computer-readable storage medium of claim 9, wherein the instructions, when executed, further cause the apparatus to: identify a gaze direction of the eye of the first user; compare the gaze direction of the eye of the first user with a corresponding location defined by the digital object; determine the focus location based upon a correlation between the gaze direction and the corresponding location of the digital object; and determine the concern of the first user based on the focus location and the gaze direction.

11. The non-transitory computer-readable storage medium of claim 9, wherein the instructions,

when executed, further cause the apparatus to augment at least a first portion of content of the one or more portions of content of the digital object based on the focus location and the concern of the first user, wherein augmenting the first portion of content of the digital object comprises one or more of (i) visually indicating the first portion of the content of the digital object corresponding to the focus location, (ii) removing the first portion of the content of the digital object corresponding to the focus location, (iii) replacing the first portion of the content of the digital object corresponding to the focus location, or (iv) generating a new digital object that does not include the first portion of content of the digital object corresponding to the focus location.

12. The non-transitory computer-readable storage medium of claim 9, wherein the one or more user parameters are associated with a user profile of the first user, and wherein the one or more user parameters comprise at least one or more of the account information, one or more user preferences, one or more concerns of the first user, one or more user identifiers, one or more user device identifiers, or transaction information associated with the first user.
